

Tutorial Worksheet 1 - Extrema of functions and Convexity

Problem 1. (Functions of Two Variables)

(a) Find the local maximum and minimum of $f(x, y) = 2x^2 - xy + 2y^2 - 20x$.

Solution: The first-order partial derivatives are

$$f_x(x, y) = 4x - y - 20,$$

and

$$f_y(x, y) = -x + 4y.$$

For a point of extremum, we require

$$f_x(x, y) = 0, \quad f_y(x, y) = 0.$$

Therefore,

$$\begin{cases} 4x - y = 20, \\ -x + 4y = 0. \end{cases}$$

From the second equation,

$$x = 4y.$$

Substituting into the first equation gives

$$4(4y) - y = 20,$$

so that

$$15y = 20 \implies y = \frac{4}{3}.$$

Hence,

$$x = 4y = \frac{16}{3}.$$

Thus, the only critical point is

$$\boxed{\left(\frac{16}{3}, \frac{4}{3}\right)}.$$

Now, the second-order partial derivatives are

$$f_{xx}(x, y) = 4, \quad f_{yy}(x, y) = 4, \quad f_{xy}(x, y) = -1.$$

The Hessian matrix is

$$H_f(x, y) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ -1 & 4 \end{bmatrix}.$$

At the critical point,

$$f_{xx}\left(\frac{16}{3}, \frac{4}{3}\right) = 4 > 0,$$

and

$$\det(H_f) = f_{xx}f_{yy} - f_{xy}^2 = (4)(4) - (-1)^2 = 16 - 1 = 15 > 0.$$

Hence, the Hessian matrix is positive definite. Therefore,

$$\boxed{\left(\frac{16}{3}, \frac{4}{3}\right)}$$

is a strict local minimum point.

The corresponding minimum value is

$$\begin{aligned} f\left(\frac{16}{3}, \frac{4}{3}\right) &= 2\left(\frac{16}{3}\right)^2 - \left(\frac{16}{3}\right)\left(\frac{4}{3}\right) + 2\left(\frac{4}{3}\right)^2 - 20\left(\frac{16}{3}\right) \\ &= -\frac{160}{3}. \end{aligned}$$

Therefore,

$$\boxed{f_{\min} = -\frac{160}{3}}.$$

There is no local maximum.

(b) Given n points (x_i, y_i) , where the x_i 's are distinct, find a and b such that the function

$$f(a, b) = \sum_{i=1}^n (ax_i + b - y_i)^2$$

is minimum.

Solution: We have

$$f_a = \frac{\partial f}{\partial a} = \sum_{i=1}^n 2(ax_i + b - y_i)x_i,$$

and

$$f_b = \frac{\partial f}{\partial b} = \sum_{i=1}^n 2(ax_i + b - y_i).$$

Also,

$$f_{aa} = 2 \sum_{i=1}^n x_i^2, \quad f_{bb} = 2n, \quad f_{ab} = 2 \sum_{i=1}^n x_i.$$

For a point of extremum,

$$f_a = 0, \quad f_b = 0.$$

Thus,

$$\sum_{i=1}^n (ax_i + b - y_i)x_i = 0,$$

and

$$\sum_{i=1}^n (ax_i + b - y_i) = 0.$$

Solving these equations gives

$$\boxed{\hat{a} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}}$$

and

$$\boxed{\hat{b} = \bar{y} - \hat{a}\bar{x}}.$$

Now, the Hessian matrix is

$$H_f(a, b) = \begin{bmatrix} f_{aa} & f_{ab} \\ f_{ab} & f_{bb} \end{bmatrix} = 2 \begin{bmatrix} \sum_{i=1}^n x_i^2 & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & n \end{bmatrix}.$$

Moreover,

$$f_{aa}(\hat{a}, \hat{b}) = 2 \sum_{i=1}^n x_i^2 > 0,$$

and

$$\begin{aligned} \det(H_f) &= 4 \left[n \sum_{i=1}^n x_i^2 - \left(\sum_{i=1}^n x_i \right)^2 \right] \\ &= 4n \sum_{i=1}^n (x_i - \bar{x})^2 > 0. \end{aligned}$$

Hence, the Hessian is positive definite. Therefore,

$$\boxed{f(\hat{a}, \hat{b})}$$

is the minimum value of $f(a, b)$.

(c) Show that $f(x, y) = y^2 + x^2y + x^4$ has a minimum at $(0, 0)$.

Solution: For a point of extremum, we require

$$f_x(x, y) = 0, \quad f_y(x, y) = 0.$$

Now,

$$f_x(x, y) = 2xy + 4x^3,$$

and

$$f_y(x, y) = 2y + x^2.$$

Therefore,

$$2xy + 4x^3 = 0, \quad 2y + x^2 = 0.$$

Clearly,

$$(x, y) = (0, 0)$$

is a critical point.

Now, the second-order partial derivatives are

$$f_{xx}(x, y) = 2y + 12x^2,$$

$$f_{xy}(x, y) = 2x,$$

and

$$f_{yy}(x, y) = 2.$$

Thus, the Hessian matrix at $(0, 0)$ is

$$H_f(0, 0) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}_{(0,0)} = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}.$$

Hence,

$$\det H_f(0, 0) = f_{xx}(0, 0)f_{yy}(0, 0) - (f_{xy}(0, 0))^2 = 0.$$

Therefore, the second derivative test is inconclusive, and further investigation is required.

Observe that

$$\begin{aligned} f(x, y) &= y^2 + x^2y + x^4 \\ &= \left(y^2 + 2y \frac{x^2}{2} + \frac{x^4}{4} \right) + \frac{3x^4}{4} \\ &= \left(y + \frac{x^2}{2} \right)^2 + \frac{3x^4}{4}. \end{aligned}$$

Therefore,

$$f(x, y) = \left(y + \frac{x^2}{2}\right)^2 + \frac{3x^4}{4} \geq 0.$$

Moreover,

$$f(x, y) = 0$$

if and only if

$$y + \frac{x^2}{2} = 0 \quad \text{and} \quad \frac{3x^4}{4} = 0.$$

The second condition implies

$$x = 0,$$

and hence the first condition gives

$$y = 0.$$

Thus,

$$f(x, y) \geq f(0, 0) = 0$$

for all (x, y) , with equality only at $(0, 0)$.

Hence, $f(x, y)$ has a strict global minimum at

$$\boxed{(0, 0)},$$

and the minimum value is

$$\boxed{f_{\min} = 0}.$$

(d) Show that the function $f(x, y) = 2x^4 - 3x^2y + y^2$ has neither a maximum nor a minimum at $(0, 0)$.

Solution: The first-order partial derivatives are

$$f_x(x, y) = 8x^3 - 6xy,$$

and

$$f_y(x, y) = -3x^2 + 2y.$$

Also,

$$f_{xx}(x, y) = 24x^2 - 6y, \quad f_{yy}(x, y) = 2, \quad f_{xy}(x, y) = -6x.$$

Now,

$$f_x(0, 0) = 0, \quad f_y(0, 0) = 0.$$

Thus, $(0, 0)$ is a critical point.

Moreover,

$$f_{xx}(0, 0)f_{yy}(0, 0) - (f_{xy}(0, 0))^2 = 0.$$

Hence, the second derivative test is inconclusive and further investigation is required.

Now,

$$\begin{aligned} f(x, y) - f(0, 0) &= 2x^4 - 3x^2y + y^2 \\ &= (y - x^2)(y - 2x^2). \end{aligned}$$

Therefore,

$$f(x, y) - f(0, 0) \begin{cases} < 0, & \text{if } x^2 < y < 2x^2, \\ > 0, & \text{if } y < x^2 \text{ or } y > 2x^2. \end{cases}$$

Thus, in every neighbourhood of $(0, 0)$, there are points for which

$$f(x, y) < f(0, 0)$$

and also points for which

$$f(x, y) > f(0, 0).$$

Hence, $f(x, y)$ has neither a local maximum nor a local minimum at $(0, 0)$. Therefore,

$$\boxed{(0, 0) \text{ is a saddle point of } f.}$$

(e) Find all local maxima and minima of $f(x, y) = xye^{-x}$.

Solution: We have

$$f(x, y) = xye^{-x}.$$

The first-order partial derivatives are

$$\begin{aligned} f_x(x, y) &= ye^{-x} - xye^{-x} \\ &= ye^{-x}(1 - x), \end{aligned}$$

and

$$f_y(x, y) = xe^{-x}.$$

For a point of extremum,

$$f_x(x, y) = 0, \quad f_y(x, y) = 0.$$

Thus,

$$ye^{-x}(1 - x) = 0, \quad xe^{-x} = 0.$$

Since

$$e^{-x} > 0$$

for every x , the second equation gives

$$x = 0.$$

Substituting $x = 0$ into the first equation gives

$$y = 0.$$

Hence, the only critical point is

$$\boxed{(x, y) = (0, 0)}.$$

Now,

$$\begin{aligned} f_{xx}(x, y) &= -ye^{-x}(1 - x) - ye^{-x} \\ &= -ye^{-x}(2 - x), \end{aligned}$$

while

$$f_{yy}(x, y) = 0,$$

and

$$f_{xy}(x, y) = e^{-x} - xe^{-x} = e^{-x}(1 - x).$$

Therefore, at $(0, 0)$,

$$f_{xx}(0, 0) = 0, \quad f_{yy}(0, 0) = 0, \quad f_{xy}(0, 0) = 1.$$

Hence,

$$\begin{aligned} \det H_f(0, 0) &= f_{xx}(0, 0)f_{yy}(0, 0) - (f_{xy}(0, 0))^2 \\ &= 0 - 1 \\ &= -1 < 0. \end{aligned}$$

Thus, the Hessian matrix

$$H_f(0, 0) = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}_{(0,0)} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is indefinite.

Therefore, $(0, 0)$ is a saddle point. Hence, the function has

$$\boxed{\text{no local maximum and no local minimum.}}$$

Problem 2. (Function of Several Variables)

(a) Find all the local maxima and minima of $f(x, y, z) = x^2 + 2y^2 + z^2 - 2xy + yz$.

Solution: The first-order partial derivatives are

$$f_x = 2x - 2y,$$

$$f_y = 4y - 2x + z,$$

and

$$f_z = 2z + y.$$

For a point of extremum, we require

$$f_x = 0, \quad f_y = 0, \quad f_z = 0.$$

Therefore,

$$\begin{cases} 2x - 2y = 0, \\ -2x + 4y + z = 0, \\ y + 2z = 0. \end{cases}$$

Equivalently,

$$\begin{bmatrix} 2 & -2 & 0 \\ -2 & 4 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Now,

$$\det \begin{bmatrix} 2 & -2 & 0 \\ -2 & 4 & 1 \\ 0 & 1 & 2 \end{bmatrix} = 6 \neq 0.$$

Hence, the coefficient matrix is nonsingular, and therefore the homogeneous system has the unique solution

$$\boxed{(x, y, z) = (0, 0, 0)}.$$

Thus, $(0, 0, 0)$ is the only critical point.

Now, the second-order partial derivatives are

$$f_{xx} = 2, \quad f_{yy} = 4, \quad f_{zz} = 2,$$

$$f_{xy} = f_{yx} = -2, \quad f_{yz} = f_{zy} = 1, \quad f_{xz} = f_{zx} = 0.$$

Hence, the Hessian matrix is

$$H_f = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix} = \begin{bmatrix} 2 & -2 & 0 \\ -2 & 4 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

To check positive definiteness, consider the leading principal minors:

$$\Delta_1 = 2 > 0,$$

$$\Delta_2 = \begin{vmatrix} 2 & -2 \\ -2 & 4 \end{vmatrix} = 8 - 4 = 4 > 0,$$

and

$$\Delta_3 = \det(H_f) = 6 > 0.$$

Therefore, by Sylvester's criterion,

H_f is positive definite.

Hence, $(0, 0, 0)$ is a strict local minimum point.

Moreover,

$$f(0, 0, 0) = 0.$$

Therefore,

$$\boxed{f_{\min} = 0 \quad \text{at} \quad (x, y, z) = (0, 0, 0).}$$

There is no local maximum.

(b) Show that the function $f(x, y, z) = x^4 + y^4 + z^4 - 4xyz$ has a minimum at $(1, 1, 1)$.

Solution: The first-order partial derivatives are

$$f_x = 4x^3 - 4yz, \quad f_y = 4y^3 - 4xz, \quad f_z = 4z^3 - 4xy.$$

For a point of extremum,

$$f_x = f_y = f_z = 0.$$

Hence,

$$x^3 = yz, \quad y^3 = xz, \quad z^3 = xy.$$

In particular, $(1, 1, 1)$ satisfies these equations. Therefore,

$$(1, 1, 1)$$

is a critical point.

Now, the second-order partial derivatives are

$$f_{xx} = 12x^2, \quad f_{yy} = 12y^2, \quad f_{zz} = 12z^2,$$

and

$$f_{xy} = -4z, \quad f_{xz} = -4y, \quad f_{yz} = -4x.$$

Thus, the Hessian matrix is

$$H_f(x, y, z) = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}.$$

At $(1, 1, 1)$,

$$H_f(1, 1, 1) = \begin{bmatrix} 12 & -4 & -4 \\ -4 & 12 & -4 \\ -4 & -4 & 12 \end{bmatrix}.$$

To check positive definiteness, consider the leading principal minors:

$$\Delta_1 = 12 > 0,$$

$$\Delta_2 = \begin{vmatrix} 12 & -4 \\ -4 & 12 \end{vmatrix} = 144 - 16 = 128 > 0,$$

and

$$\Delta_3 = \det \begin{bmatrix} 12 & -4 & -4 \\ -4 & 12 & -4 \\ -4 & -4 & 12 \end{bmatrix} = 1024 > 0.$$

Hence, by Sylvester's criterion, the Hessian matrix is positive definite at $(1, 1, 1)$.

Therefore,

$$\boxed{(1, 1, 1)}$$

is a strict local minimum point of f .

The corresponding minimum value is

$$f(1, 1, 1) = 1 + 1 + 1 - 4 = -1.$$

Thus,

$$\boxed{f(1, 1, 1) = -1}.$$

Problem 3. (Constrained Optimization with Lagrange Multiplier)

- (a) Find all the local maxima and minima of $f(x, y) = 9 - (x - 1)^2 - y^2$ subject to $x^2 + y^2 = 4$.

Solution: Let

$$z = f(x, y)$$

and define the constraint function

$$g(x, y) = x^2 + y^2 - 4.$$

We wish to maximize or minimize

$$z = f(x, y)$$

subject to

$$g(x, y) = 0.$$

Using Lagrange Multiplier method:

Define

$$F(x, y) = f(x, y) + \lambda g(x, y).$$

Thus,

$$F(x, y) = 9 - (x - 1)^2 - y^2 + \lambda(x^2 + y^2 - 4).$$

For a point of extremum,

$$\frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0, \quad g(x, y) = 0.$$

Now,

$$\frac{\partial F}{\partial x} = -2(x - 1) + 2\lambda x,$$

and

$$\frac{\partial F}{\partial y} = -2y + 2\lambda y.$$

Therefore,

$$\begin{cases} -2(x - 1) + 2\lambda x = 0, \\ -2y + 2\lambda y = 0, \\ x^2 + y^2 = 4. \end{cases}$$

Equivalently,

$$\begin{cases} (1 - \lambda)x = 1, \\ (1 - \lambda)y = 0, \\ x^2 + y^2 = 4. \end{cases}$$

From (1), we must have

$$\lambda \neq 1.$$

Hence, from (2),

$$y = 0.$$

Also, from (1),

$$x = \frac{1}{1 - \lambda}.$$

Using the constraint,

$$x^2 + y^2 = 4,$$

we obtain

$$\left(\frac{1}{1 - \lambda} \right)^2 = 4.$$

Hence,

$$\lambda = \frac{1}{2} \quad \text{or} \quad \lambda = \frac{3}{2}.$$

Therefore,

$$(x, y) = (2, 0) \quad \text{or} \quad (x, y) = (-2, 0).$$

Since

$$x^2 + y^2 = 4$$

is a closed and bounded curve, the absolute maximum and minimum of $f(x, y)$ exist on the constraint.

Now,

$$f(2, 0) = 9 - (2 - 1)^2 = 8,$$

whereas

$$f(-2, 0) = 9 - (-2 - 1)^2 = 0.$$

Therefore, the constrained maximum is

$$\boxed{f_{\max} = 8}$$

attained at

$$\boxed{(2, 0)},$$

and the constrained minimum is

$$\boxed{f_{\min} = 0}$$

attained at

$$\boxed{(-2, 0)}.$$

- (b) Show that the volume of the greatest rectangular parallelepiped (a three-dimensional geometric shape with six faces, where every face is a parallelogram) that can be inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ is $\frac{abc}{3\sqrt{3}}$.

Solution: The volume of the parallelepiped is

$$V = xyz.$$

Thus, we want to maximize

$$V = xyz$$

subject to the constraint

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, \quad x, y, z > 0.$$

Since the logarithm is a strictly increasing function, maximizing $V = xyz$ is equivalent to maximizing

$$\ln V = \ln(xyz).$$

Using the method of Lagrange multipliers, define

$$F(x, y, z) = \ln(xyz) + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right).$$

For a point of extremum,

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial y} = \frac{\partial F}{\partial z} = 0.$$

Therefore,

$$\frac{1}{x} + \frac{2\lambda x}{a^2} = 0,$$

$$\frac{1}{y} + \frac{2\lambda y}{b^2} = 0,$$

and

$$\frac{1}{z} + \frac{2\lambda z}{c^2} = 0.$$

Hence,

$$x^2 = -\frac{a^2}{2\lambda}, \quad y^2 = -\frac{b^2}{2\lambda}, \quad z^2 = -\frac{c^2}{2\lambda}.$$

Substituting these into the constraint,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1,$$

we obtain

$$-\frac{1}{2\lambda} - \frac{1}{2\lambda} - \frac{1}{2\lambda} = 1.$$

Thus,

$$-\frac{3}{2\lambda} = 1,$$

and hence

$$\lambda = -\frac{3}{2}.$$

Therefore,

$$x^2 = \frac{a^2}{3}, \quad y^2 = \frac{b^2}{3}, \quad z^2 = \frac{c^2}{3}.$$

Since $x, y, z > 0$,

$$(x, y, z) = \left(\frac{a}{\sqrt{3}}, \frac{b}{\sqrt{3}}, \frac{c}{\sqrt{3}} \right).$$

Hence,

$$V_{\max} = \frac{a}{\sqrt{3}} \cdot \frac{b}{\sqrt{3}} \cdot \frac{c}{\sqrt{3}} = \boxed{\frac{abc}{3\sqrt{3}}}.$$

Thus, the maximum volume is

$$\boxed{V_{\max} = \frac{abc}{3\sqrt{3}}}.$$

- (c) If $ax^2 + by^2 = ab$, show that the maximum and minimum values of $x^2 + xy + y^2$ are the values of θ given by $4(\theta - a)(\theta - b) = ab$.

Solution: Let

$$F(x, y) = x^2 + xy + y^2 + \lambda(ax^2 + by^2 - ab).$$

For a point of extremum, we require

$$\frac{\partial F}{\partial x} = 0, \quad \frac{\partial F}{\partial y} = 0,$$

together with the constraint

$$ax^2 + by^2 = ab.$$

Now,

$$\frac{\partial F}{\partial x} = 2x + y + 2\lambda ax = 0, \tag{1}$$

and

$$\frac{\partial F}{\partial y} = x + 2y + 2\lambda by = 0. \tag{2}$$

Suppose that at an extremum,

$$\theta = x^2 + xy + y^2.$$

Multiplying (1) by x and (2) by y , and then adding, we get

$$2x^2 + 2xy + 2y^2 + 2\lambda(ax^2 + by^2) = 0.$$

Using

$$x^2 + xy + y^2 = \theta$$

and

$$ax^2 + by^2 = ab,$$

we obtain

$$2\theta + 2\lambda ab = 0.$$

Hence,

$$\boxed{\lambda = -\frac{\theta}{ab}}.$$

Also, equations (1) and (2) can be written as

$$2(1 + \lambda a)x + y = 0,$$

and

$$x + 2(1 + \lambda b)y = 0.$$

Since $(x, y) \neq (0, 0)$, these equations must have a non-trivial solution. Therefore, the determinant of the coefficient matrix must vanish:

$$\begin{vmatrix} 2(1 + \lambda a) & 1 \\ 1 & 2(1 + \lambda b) \end{vmatrix} = 0.$$

Thus,

$$4(1 + \lambda a)(1 + \lambda b) - 1 = 0.$$

Substituting

$$\lambda = -\frac{\theta}{ab},$$

we get

$$4\left(1 - \frac{\theta}{b}\right)\left(1 - \frac{\theta}{a}\right) = 1.$$

Therefore,

$$4\frac{(b - \theta)(a - \theta)}{ab} = 1,$$

which gives

$$4(a - \theta)(b - \theta) = ab.$$

Hence,

$$\boxed{4(\theta - a)(\theta - b) = ab}.$$

Thus, the maximum and minimum values of

$$x^2 + xy + y^2$$

subject to

$$ax^2 + by^2 = ab$$

are precisely the two roots of

$$\boxed{4(\theta - a)(\theta - b) = ab}.$$

Solving the quadratic gives

$$\theta_{\max, \min} = \frac{a + b \pm \sqrt{a^2 - ab + b^2}}{2}.$$

(d) Given m positive numbers $c_1, c_2, \dots, c_m$, find the maximum value of

$$\sum_{i=1}^m c_i x_i,$$

where the variables x_i are restricted so that

$$\sum_{i=1}^m x_i^2 = 1.$$

Solution: We wish to maximize

$$f(\mathbf{x}) = \sum_{i=1}^m c_i x_i$$

subject to the constraint

$$g(\mathbf{x}) = \sum_{i=1}^m x_i^2 - 1 = 0.$$

Define

$$F(\mathbf{x}) = f(\mathbf{x}) + \lambda g(\mathbf{x}),$$

that is,

$$F(\mathbf{x}) = \sum_{i=1}^m c_i x_i + \lambda \left(\sum_{i=1}^m x_i^2 - 1 \right).$$

For a point of extremum,

$$\frac{\partial F}{\partial x_i} = 0, \quad i = 1, 2, \dots, m,$$

together with

$$\sum_{i=1}^m x_i^2 = 1.$$

Thus,

$$c_i + 2\lambda x_i = 0, \quad i = 1, 2, \dots, m.$$

Hence,

$$x_i = -\frac{c_i}{2\lambda}.$$

Using the constraint,

$$1 = \sum_{i=1}^m x_i^2,$$

we obtain

$$1 = \frac{1}{4\lambda^2} \sum_{i=1}^m c_i^2.$$

Therefore,

$$4\lambda^2 = \sum_{i=1}^m c_i^2,$$

or

$$2\lambda = \pm \sqrt{\sum_{i=1}^m c_i^2}.$$

Hence,

$$x_i = \pm \frac{c_i}{\sqrt{\sum_{j=1}^m c_j^2}}, \quad i = 1, 2, \dots, m.$$

Since

$$\sum_{i=1}^m x_i^2 = 1$$

is a closed and bounded set, the absolute maximum and minimum exist.

For

$$x_i = \frac{c_i}{\sqrt{\sum_{j=1}^m c_j^2}},$$

we have

$$\sum_{i=1}^m c_i x_i = \sum_{i=1}^m c_i \frac{c_i}{\sqrt{\sum_{j=1}^m c_j^2}} = \frac{\sum_{i=1}^m c_i^2}{\sqrt{\sum_{i=1}^m c_i^2}} = \sqrt{\sum_{i=1}^m c_i^2}.$$

Similarly, for

$$x_i = -\frac{c_i}{\sqrt{\sum_{j=1}^m c_j^2}},$$

we obtain

$$\sum_{i=1}^m c_i x_i = -\sqrt{\sum_{i=1}^m c_i^2}.$$

Therefore,

$$\max \sum_{i=1}^m c_i x_i = \sqrt{\sum_{i=1}^m c_i^2}$$

and

$$\min \sum_{i=1}^m c_i x_i = -\sqrt{\sum_{i=1}^m c_i^2}.$$

Remark.

The same result follows immediately from the Cauchy-Schwarz inequality:

$$\left(\sum_{i=1}^m c_i x_i \right)^2 \leq \left(\sum_{i=1}^m c_i^2 \right) \left(\sum_{i=1}^m x_i^2 \right).$$

Since

$$\sum_{i=1}^m x_i^2 = 1,$$

we get

$$\left| \sum_{i=1}^m c_i x_i \right| \leq \sqrt{\sum_{i=1}^m c_i^2}.$$

Equivalently,

$$-\sqrt{\sum_{i=1}^m c_i^2} \leq \sum_{i=1}^m c_i x_i \leq \sqrt{\sum_{i=1}^m c_i^2}.$$

(e) Find the minimum value of

$$\sum_{i=1}^n a_i^2$$

subject to

$$\sum_{i=1}^n a_i = 1.$$

Solution: Using the Cauchy–Schwarz inequality,

$$\left(\sum_{i=1}^n a_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n 1^2\right).$$

Since

$$\sum_{i=1}^n a_i = 1$$

and

$$\sum_{i=1}^n 1^2 = n,$$

we obtain

$$1 \leq n \sum_{i=1}^n a_i^2.$$

Hence,

$$\sum_{i=1}^n a_i^2 \geq \frac{1}{n}.$$

Equality holds in the Cauchy–Schwarz inequality if

$$a_i = k \cdot 1 \quad \text{for all } i,$$

i.e.,

$$a_1 = a_2 = \cdots = a_n = k.$$

Since

$$\sum_{i=1}^n a_i = 1,$$

we get

$$nk = 1,$$

and therefore

$$a_i = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

Thus,

$$\boxed{\min \sum_{i=1}^n a_i^2 = \frac{1}{n}}$$

and the minimum occurs when

$$\boxed{a_1 = a_2 = \cdots = a_n = \frac{1}{n}.$$

Using Lagrange multipliers:

Define

$$F(a_1, \dots, a_n) = \sum_{i=1}^n a_i^2 + \lambda \left(\sum_{i=1}^n a_i - 1 \right).$$

For a point of extremum,

$$\frac{\partial F}{\partial a_i} = 0, \quad i = 1, 2, \dots, n,$$

together with

$$\sum_{i=1}^n a_i = 1.$$

Now,

$$\frac{\partial F}{\partial a_i} = 2a_i + \lambda.$$

Hence,

$$2a_i + \lambda = 0, \quad i = 1, 2, \dots, n.$$

Therefore,

$$a_i = -\frac{\lambda}{2}.$$

Using the constraint,

$$1 = \sum_{i=1}^n a_i = n \left(-\frac{\lambda}{2} \right),$$

so

$$\lambda = -\frac{2}{n}.$$

Thus,

$$a_i = \frac{1}{n}, \quad i = 1, 2, \dots, n.$$

Finally, note that

$$\begin{aligned} \sum_{i=1}^n a_i^2 &= \sum_{i=1}^n \left(a_i - \frac{1}{n} + \frac{1}{n} \right)^2 \\ &= \sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{2}{n} \sum_{i=1}^n \left(a_i - \frac{1}{n} \right) + \frac{n}{n^2}. \end{aligned}$$

Since

$$\sum_{i=1}^n \left(a_i - \frac{1}{n} \right) = \sum_{i=1}^n a_i - 1 = 0,$$

we get

$$\sum_{i=1}^n a_i^2 = \sum_{i=1}^n \left(a_i - \frac{1}{n} \right)^2 + \frac{1}{n}.$$

Therefore,

$$\sum_{i=1}^n a_i^2 \geq \frac{1}{n},$$

with equality if and only if

$$a_i = \frac{1}{n} \quad \text{for all } i.$$

(f) Find the maximum and minimum values of

$$\mathbf{x}^\top A \mathbf{x}$$

subject to

$$\mathbf{x}^\top \mathbf{x} = 1,$$

where A is a symmetric matrix.

Solution: Define the Lagrangian

$$f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} - \lambda (\mathbf{x}^\top \mathbf{x} - 1).$$

For a point of extremum,

$$\nabla_{\mathbf{x}} f(\mathbf{x}) = \mathbf{0},$$

together with the constraint

$$\mathbf{x}^\top \mathbf{x} = 1.$$

Since A is symmetric,

$$\nabla_{\mathbf{x}} (\mathbf{x}^\top A \mathbf{x}) = 2A\mathbf{x},$$

and

$$\nabla_{\mathbf{x}} (\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}.$$

Hence,

$$2A\mathbf{x} - 2\lambda\mathbf{x} = \mathbf{0}.$$

Therefore,

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Thus, at any constrained extremum, $\mathbf{x}$ must be an eigenvector of A , and λ is the corresponding eigenvalue.

Now, premultiplying by $\mathbf{x}^\top$, we get

$$\mathbf{x}^\top A \mathbf{x} = \lambda \mathbf{x}^\top \mathbf{x}.$$

Since

$$\mathbf{x}^\top \mathbf{x} = 1,$$

it follows that

$$\mathbf{x}^\top A \mathbf{x} = \lambda.$$

Therefore, the maximum value is the largest eigenvalue of A , while the minimum value is the smallest eigenvalue of A .

If the eigenvalues are ordered as

$$\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n,$$

then

$$\boxed{\max_{\mathbf{x}^\top \mathbf{x}=1} \mathbf{x}^\top A \mathbf{x} = \lambda_n}$$

and

$$\boxed{\min_{\mathbf{x}^\top \mathbf{x}=1} \mathbf{x}^\top A \mathbf{x} = \lambda_1}.$$

The maximum is attained at any unit eigenvector corresponding to λ_n , and the minimum is attained at any unit eigenvector corresponding to λ_1 .

Remark.

For any nonzero vector $\mathbf{x}$,

$$R_A(\mathbf{x}) = \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

is called the *Rayleigh quotient*.

Its maximum and minimum values are

$$\max_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \lambda_n$$

and

$$\min_{\mathbf{x} \neq \mathbf{0}} \frac{\mathbf{x}^\top A \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \lambda_1.$$

(g) If

$$2x + 3y + 4z = a,$$

show that the maximum value of

$$x^2 y^3 z^4$$

is

$$\left(\frac{a}{9}\right)^9.$$

Solution: We wish to maximize

$$f(x, y, z) = x^2 y^3 z^4$$

subject to the constraint

$$2x + 3y + 4z = a.$$

Using the method of Lagrange multipliers, define

$$F(x, y, z) = x^2 y^3 z^4 + \lambda(2x + 3y + 4z - a).$$

For a point of extremum,

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial y} = \frac{\partial F}{\partial z} = 0.$$

Thus,

$$\frac{\partial F}{\partial x} = 2xy^3z^4 + 2\lambda = 0,$$

$$\frac{\partial F}{\partial y} = 3x^2y^2z^4 + 3\lambda = 0,$$

and

$$\frac{\partial F}{\partial z} = 4x^2y^3z^3 + 4\lambda = 0.$$

Hence,

$$2xy^3z^4 = -2\lambda,$$

$$3x^2y^2z^4 = -3\lambda,$$

and

$$4x^2y^3z^3 = -4\lambda.$$

Dividing the first equation by the second,

$$\frac{2xy^3z^4}{3x^2y^2z^4} = \frac{2}{3},$$

which gives

$$\frac{2y}{3x} = \frac{2}{3},$$

and therefore

$$x = y.$$

Similarly, dividing the second equation by the third,

$$\frac{3x^2y^2z^4}{4x^2y^3z^3} = \frac{3}{4},$$

which gives

$$\frac{3z}{4y} = \frac{3}{4},$$

and hence

$$y = z.$$

Therefore,

$$x = y = z.$$

Using the constraint,

$$2x + 3x + 4x = a,$$

so that

$$9x = a.$$

Thus,

$$\boxed{x = y = z = \frac{a}{9}}.$$

Therefore,

$$x^2y^3z^4 = \left(\frac{a}{9}\right)^2 \left(\frac{a}{9}\right)^3 \left(\frac{a}{9}\right)^4 = \left(\frac{a}{9}\right)^9.$$

Hence,

$$\boxed{\max x^2y^3z^4 = \left(\frac{a}{9}\right)^9}.$$

Alternative proof using the AM–GM inequality.

Assuming $x, y, z > 0$, apply AM–GM to the nine positive numbers

$$x, x, y, y, y, z, z, z, z.$$

Then

$$\frac{2x + 3y + 4z}{9} \geq (x^2y^3z^4)^{1/9}.$$

Since

$$2x + 3y + 4z = a,$$

we obtain

$$\frac{a}{9} \geq (x^2y^3z^4)^{1/9}.$$

Raising both sides to the ninth power,

$$x^2y^3z^4 \leq \left(\frac{a}{9}\right)^9.$$

Equality in AM–GM holds when

$$x = y = z = \frac{a}{9}.$$

Hence,

$$\boxed{\max x^2y^3z^4 = \left(\frac{a}{9}\right)^9}.$$

Problem 4. (Geometry of Linear and Quadratic Programming)

(a) Consider the optimization problem

$$\begin{aligned} \max_{x,y} \quad & 4x + y \\ \text{s.t.} \quad & 3x + 2y \leq 18, \\ & -x + 2y \geq 2, \\ & x \leq 3, \\ & x, y \geq 0. \end{aligned}$$

(i) Find the optimal solution when $x, y \in \mathbb{R}$ (Continuous Linear Programming).

Solution: Rewrite the constraints as

$$\begin{aligned} y &\leq 9 - \frac{3}{2}x \\ y &\geq 1 + \frac{x}{2} \\ 0 &\leq x \leq 3 \end{aligned}$$

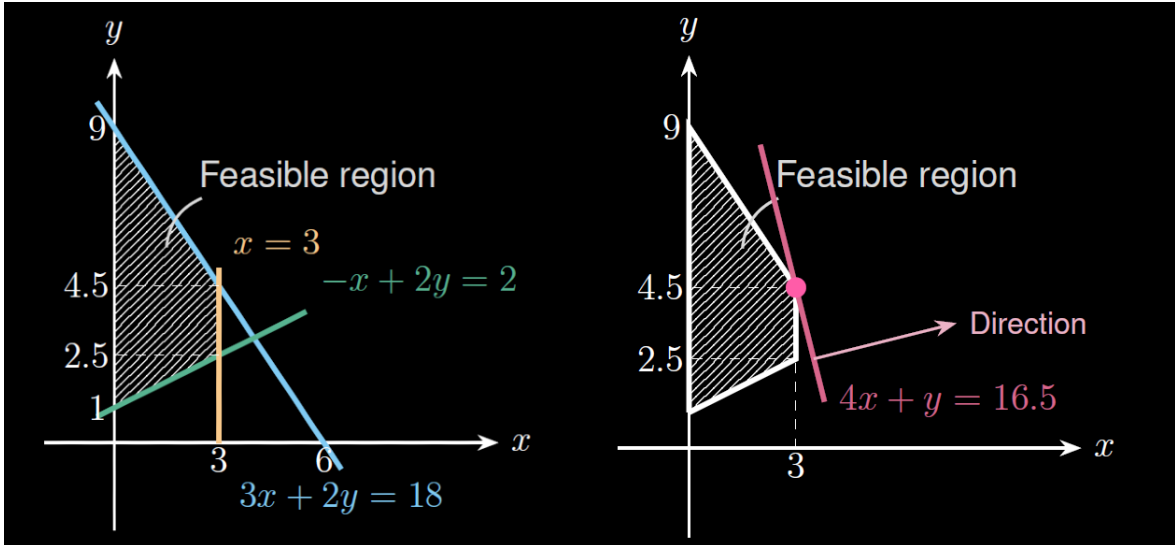


Figure 1: Plot

The feasible region has four vertices:

$$(0, 1), \quad (0, 9), \quad \left(3, \frac{9}{2}\right), \quad \left(3, \frac{5}{2}\right).$$

Evaluate

$$f(x, y) = 4x + y$$

(x, y)	$4x + y$
$(0, 1)$	1
$(0, 9)$	9
$\left(3, \frac{9}{2}\right)$	$\frac{33}{2} = 16.5$
$\left(3, \frac{5}{2}\right)$	$\frac{29}{2} = 14.5$

Therefore,

$$x^* = 3, \quad y^* = \frac{9}{2}$$

and

$$f^* = \frac{33}{2} = 16.5$$

Geometrically, the objective lines are

$$4x + y = c \iff y = -4x + c.$$

They all have slope -4. Moving this line in the direction of increasing c , the final point of contact with the feasible region is

$$\left(3, \frac{9}{2}\right)$$

(ii) Find the optimal solution when $x, y \in \mathbb{Z}$ (Integer Linear Programming).

Solution:

$$x, y \in \mathbb{Z}.$$

Since

$$0 \leq x \leq 3,$$

we only need to consider

$$x = 0, 1, 2, 3.$$

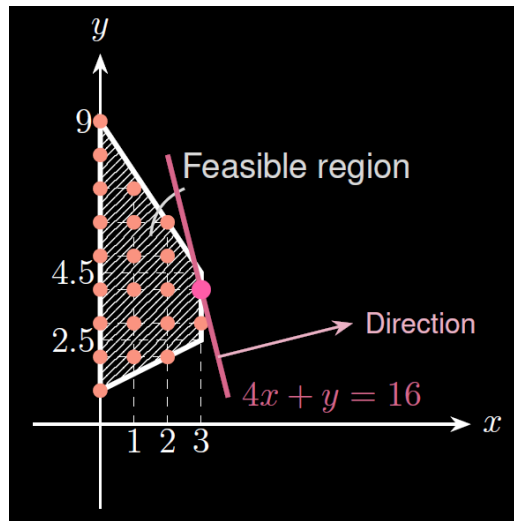


Figure 2: Plot

Because the objective coefficient of y is positive, for each x we choose the largest feasible integer y .

x	largest feasible integer y	$4x + y$
0	9	9
1	7	11
2	6	14
3	4	16

Hence,

$$(x^*, y^*) = (3, 4)$$

with

$$f^* = 16.$$

(iii) Find the optimal solution when $x \in \mathbb{Z}$, $y \in \mathbb{R}$ (Mixed-Integer Programming).

Solution:

$$x \in \mathbb{Z}, y \in \mathbb{R}$$

Again,

$$x \in \{0, 1, 2, 3\}.$$

For each x , maximize y , so the upper constraint

$$3x + 2y \leq 18$$

gives

$$y = 9 - \frac{3}{2}x.$$

Therefore,

$$4x + y = 4x + 9 - \frac{3}{2}x = 9 + \frac{5}{2}x.$$

This increases with x , so choose $x = 3$.

Then

$$y = 9 - \frac{3}{2}(3) = \frac{9}{2}.$$

Hence,

$$(x^*, y^*) = \left(3, \frac{9}{2}\right), \quad f^* = 16.5.$$

Interestingly, imposing integrality on x alone makes no difference, because the continuous LP solution already has

$$x^* = 3 \in \mathbb{Z}.$$

(iv) Find the optimal solution when $x \in \{0, 1\}$, $y \in \mathbb{R}$ (Binary and Mixed-Integer Decisions).

Solution: $x \in \{0, 1\}, y \in \mathbb{R}$ For $x = 0$,

$$y \leq 9,$$

so

$$f = 9.$$

For $x = 1$,

$$y \leq \frac{15}{2} = 7.5.$$

Therefore,

$$f = 4 + 7.5 = 11.5.$$

Hence,

$$(x^*, y^*) = (1, 7.5)$$

with

$$f^* = 11.5$$

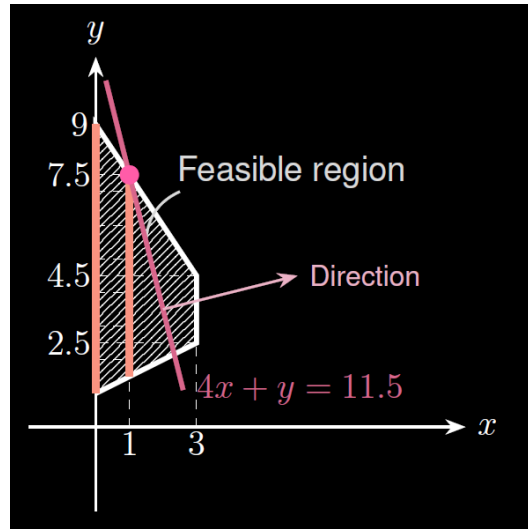


Figure 3: Plot

(b) Consider

$$\begin{aligned} \max_{x,y} \quad & 9x^2 + y^2 \\ \text{s.t.} \quad & 3x + 2y \leq 18, \\ & -x + 2y \geq 2, \\ & x \leq 3, \\ & x, y \geq 0. \end{aligned}$$

Find the optimal value when $x \in \mathbb{R}, y \in \mathbb{R}$ and $x \in \{0, 1\}, y \in \mathbb{R}$.

Solution: The constraints can be written as

$$\begin{aligned} y &\leq 9 - \frac{3}{2}x \\ y &\geq 1 + \frac{1}{2}x \end{aligned}$$

and

$$0 \leq x \leq 3.$$

Hence, the feasible region has vertices

$$(0, 1), \quad (0, 9), \quad \left(3, \frac{9}{2}\right), \quad \left(3, \frac{5}{2}\right).$$

The objective function is

$$f(x, y) = 9x^2 + y^2.$$

Evaluating it at the vertices,

(x, y)	$9x^2 + y^2$
$(0, 1)$	1
$(0, 9)$	81
$\left(3, \frac{9}{2}\right)$	$81 + \frac{81}{4} = \frac{405}{4} = 101.25$
$\left(3, \frac{5}{2}\right)$	$81 + \frac{25}{4} = \frac{349}{4} = 87.25$

Therefore,

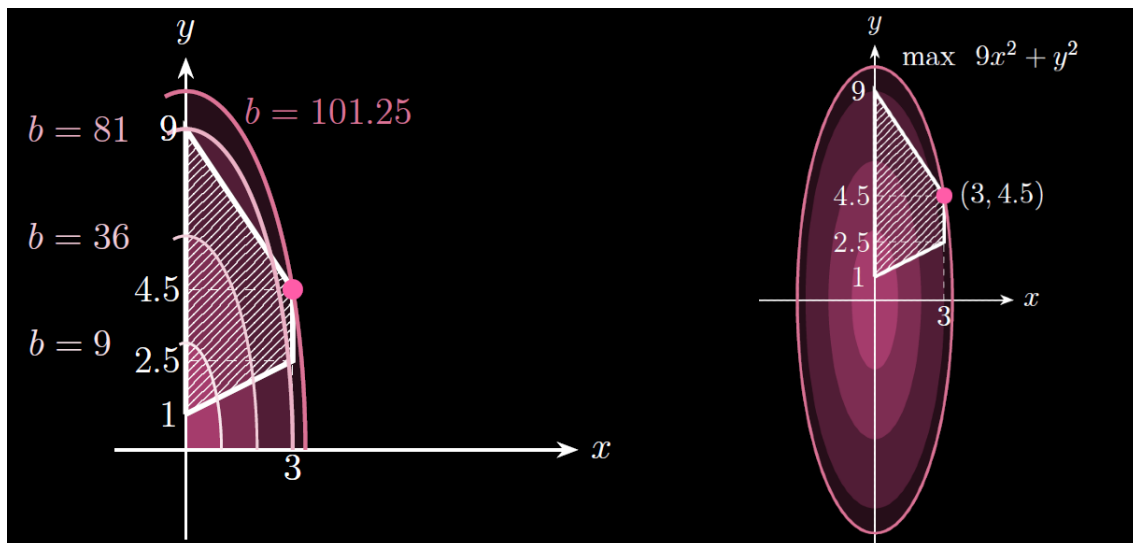


Figure 4: Plot

$$(x^*, y^*) = \left(3, \frac{9}{2}\right)$$

and

$$f^* = \frac{405}{4} = 101.25.$$

Geometrical interpretation

The level curves of the objective are

$$9x^2 + y^2 = c.$$

These are ellipses centered at the origin:

$$\frac{x^2}{c/9} + \frac{y^2}{c} = 1.$$

As c increases, the ellipse expands outward. The largest ellipse that still touches the feasible region touches it at

$$\left(3, \frac{9}{2}\right)$$

This gives the maximum.

Let's consider the second part.

Case 1: $x = 0$ The constraints become

$$2y \leq 18$$

and

$$2y \geq 2.$$

Hence,

$$1 \leq y \leq 9.$$

The objective becomes

$$f(0, y) = y^2.$$

Since $y \geq 0$, this is maximized at

$$y = 9.$$

Therefore,

$$f(0, 9) = 81.$$

Case 2: $x = 1$

The constraints become

$$3 + 2y \leq 18$$

and

$$-1 + 2y \geq 2.$$

Therefore,

$$y \leq \frac{15}{2} = 7.5$$

and

$$y \geq \frac{3}{2} = 1.5.$$

Thus,

$$\frac{3}{2} \leq y \leq \frac{15}{2}.$$

The objective becomes

$$f(1, y) = 9 + y^2.$$

This is maximized at

$$y = \frac{15}{2}.$$

Comparing the two possibilities,

$$81 > 65.25.$$

Therefore, the mixed-integer quadratic problem has solution

$$(x^*, y^*) = (0, 9)$$

with

$$f^* = 81.$$

Problem 5. (Convexity)

- (a) Let $\mathbf{x} \in \mathbb{R}^n$ and $y \in \mathbb{R}$. Consider the quadratic-over-linear function

$$f(\mathbf{x}, y) = \begin{cases} \frac{\mathbf{x}^\top \mathbf{x}}{y}, & y > 0, \\ +\infty, & y \leq 0. \end{cases}$$

- (a) Show that the effective domain of f is convex.
 (b) Compute the gradient $\nabla f(\mathbf{x}, y)$ for $y > 0$.
 (c) Compute the Hessian $\nabla^2 f(\mathbf{x}, y)$.
 (d) Show that the Hessian is positive semidefinite for every $y > 0$.
 (e) Hence, conclude that f is a convex function.

Solution: The effective domain of f is

$$\text{dom}(f) = \{(\mathbf{x}, y) \in \mathbb{R}^n \times \mathbb{R} : y > 0\}.$$

(a) Convexity of the domain.

Let

$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2) \in \text{dom}(f),$$

so that $y_1 > 0$ and $y_2 > 0$.

For any $\lambda \in [0, 1]$,

$$\lambda y_1 + (1 - \lambda)y_2 > 0.$$

Therefore,

$$\lambda(\mathbf{x}_1, y_1) + (1 - \lambda)(\mathbf{x}_2, y_2) \in \text{dom}(f),$$

and hence $\text{dom}(f)$ is convex.

(b) Gradient.

For $y > 0$,

$$f(\mathbf{x}, y) = \frac{\mathbf{x}^\top \mathbf{x}}{y}.$$

Differentiating with respect to $\mathbf{x}$ gives

$$\nabla_{\mathbf{x}} f(\mathbf{x}, y) = \frac{2\mathbf{x}}{y},$$

while

$$\frac{\partial f}{\partial y} = -\frac{\mathbf{x}^\top \mathbf{x}}{y^2}.$$

Thus,

$$\nabla f(\mathbf{x}, y) = \begin{pmatrix} \frac{2\mathbf{x}}{y} \\ -\frac{\mathbf{x}^\top \mathbf{x}}{y^2} \end{pmatrix}.$$

(c) Hessian.

Differentiating once more, we obtain

$$\nabla^2 f(\mathbf{x}, y) = \begin{pmatrix} \frac{2}{y} I & -\frac{2}{y^2} \mathbf{x} \\ -\frac{2}{y^2} \mathbf{x}^\top & \frac{2}{y^3} \mathbf{x}^\top \mathbf{x} \end{pmatrix},$$

where I is the $n \times n$ identity matrix.

(d) Positive semidefiniteness of the Hessian.

Let

$$\mathbf{v} = \begin{pmatrix} \mathbf{v}_1 \\ v_2 \end{pmatrix},$$

where

$$\mathbf{v}_1 \in \mathbb{R}^n, \quad v_2 \in \mathbb{R}.$$

Then

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x}, y) \mathbf{v} = \frac{2}{y} \|\mathbf{v}_1\|_2^2 - \frac{4v_2}{y^2} \mathbf{v}_1^\top \mathbf{x} + \frac{2v_2^2}{y^3} \mathbf{x}^\top \mathbf{x}.$$

Factoring out $\frac{2}{y^3}$,

$$\begin{aligned} \mathbf{v}^\top \nabla^2 f(\mathbf{x}, y) \mathbf{v} &= \frac{2}{y^3} (y^2 \|\mathbf{v}_1\|_2^2 - 2yv_2 \mathbf{v}_1^\top \mathbf{x} + v_2^2 \|\mathbf{x}\|_2^2) \\ &= \frac{2}{y^3} \|y\mathbf{v}_1 - v_2\mathbf{x}\|_2^2. \end{aligned}$$

Since $y > 0$,

$$\frac{2}{y^3} > 0,$$

and since a squared norm is always nonnegative,

$$\|y\mathbf{v}_1 - v_2\mathbf{x}\|_2^2 \geq 0.$$

Therefore,

$$\boxed{\mathbf{v}^\top \nabla^2 f(\mathbf{x}, y) \mathbf{v} \geq 0 \quad \text{for all } \mathbf{v}.}$$

Hence,

$$\boxed{\nabla^2 f(\mathbf{x}, y) \succeq 0.}$$

(e) Conclusion.

The domain of f is convex and its Hessian is positive semidefinite throughout the interior of its domain. Therefore,

$$\boxed{f(\mathbf{x}, y) = \frac{\mathbf{x}^\top \mathbf{x}}{y}, \quad y > 0,}$$

is a convex function.

(b) Let $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$. Consider the log-sum-exp function

$$f(\mathbf{x}) = \log \left(\sum_{i=1}^n e^{x_i} \right).$$

- Determine the domain of f and show that it is convex.
- Compute the gradient $\nabla f(\mathbf{x})$.
- Compute the Hessian $\nabla^2 f(\mathbf{x})$.
- Show that the Hessian is positive semidefinite.
- Hence, conclude that f is convex.

Solution: (a) Domain.

Since

$$e^{x_i} > 0 \quad \text{for every } x_i \in \mathbb{R},$$

we have

$$\sum_{i=1}^n e^{x_i} > 0.$$

Therefore,

$$\text{dom}(f) = \mathbb{R}^n,$$

which is a convex set.

(b) Gradient.

Let

$$S(\mathbf{x}) = \sum_{j=1}^n e^{x_j}.$$

Then

$$f(\mathbf{x}) = \log S(\mathbf{x}).$$

For each i ,

$$\frac{\partial f}{\partial x_i} = \frac{1}{S(\mathbf{x})} \frac{\partial S(\mathbf{x})}{\partial x_i} = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}.$$

Define

$$p_i = \frac{e^{x_i}}{\sum_{j=1}^n e^{x_j}}, \quad i = 1, \dots, n.$$

Then

$$p_i > 0, \quad \sum_{i=1}^n p_i = 1.$$

Hence,

$$\nabla f(\mathbf{x}) = \mathbf{p} = \begin{pmatrix} p_1 \\ \vdots \\ p_n \end{pmatrix}.$$

(c) Hessian.

We differentiate p_i with respect to x_j .

If $i = j$,

$$\frac{\partial p_i}{\partial x_i} = p_i(1 - p_i).$$

If $i \neq j$,

$$\frac{\partial p_i}{\partial x_j} = -p_i p_j.$$

Therefore,

$$\nabla^2 f(\mathbf{x}) = \text{Diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top.$$

Equivalently,

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} p_1(1-p_1) & -p_1 p_2 & \cdots & -p_1 p_n \\ -p_2 p_1 & p_2(1-p_2) & \cdots & -p_2 p_n \\ \vdots & \vdots & \ddots & \vdots \\ -p_n p_1 & -p_n p_2 & \cdots & p_n(1-p_n) \end{pmatrix}.$$

(d) Positive semidefiniteness.

Let $\mathbf{v} \in \mathbb{R}^n$. Then

$$\begin{aligned}\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} &= \mathbf{v}^\top (\text{Diag}(\mathbf{p}) - \mathbf{p} \mathbf{p}^\top) \mathbf{v} \\ &= \sum_{i=1}^n p_i v_i^2 - \left(\sum_{i=1}^n p_i v_i \right)^2.\end{aligned}$$

Let

$$\mu = \sum_{i=1}^n p_i v_i.$$

Then

$$\begin{aligned}\sum_{i=1}^n p_i (v_i - \mu)^2 &= \sum_{i=1}^n p_i v_i^2 - 2\mu \sum_{i=1}^n p_i v_i + \mu^2 \sum_{i=1}^n p_i \\ &= \sum_{i=1}^n p_i v_i^2 - 2\mu^2 + \mu^2 \\ &= \sum_{i=1}^n p_i v_i^2 - \mu^2.\end{aligned}$$

Hence,

$$\mathbf{v}^\top \nabla^2 f(\mathbf{x}) \mathbf{v} = \sum_{i=1}^n p_i (v_i - \mu)^2 \geq 0.$$

Thus,

$$\nabla^2 f(\mathbf{x}) \succeq 0.$$

(e) Conclusion.

Since the domain $\mathbb{R}^n$ is convex and the Hessian is positive semidefinite for every $\mathbf{x} \in \mathbb{R}^n$, we conclude that

$$f(\mathbf{x}) = \log \left(\sum_{i=1}^n e^{x_i} \right)$$

is a convex function.

(c) Let S_{++}^n denote the set of symmetric positive definite $n \times n$ matrices. Consider

$$f(X) = -\log \det(X), \quad X \in S_{++}^n.$$

(a) Show that S_{++}^n is a convex set.

(b) Fix $X \in S_{++}^n$ and a symmetric matrix $V \in S^n$. Define the restriction of f to the line $X + tV$ by

$$g(t) = f(X + tV) = -\log \det(X + tV),$$

for all t such that $X + tV \succ 0$.

(c) Show that

$$X + tV = X^{1/2} \left(I + tX^{-1/2} V X^{-1/2} \right) X^{1/2}.$$

(d) Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of

$$M = X^{-1/2} V X^{-1/2}.$$

Show that

$$g(t) = -\log \det X - \sum_{i=1}^n \log(1 + t\lambda_i).$$

(e) Compute $g''(t)$ and show that $g''(t) \geq 0$.

(f) Hence conclude that

$$f(X) = -\log \det X$$

is convex on S_{++}^n .

Solution: 1. Convexity of the domain.

Let

$$X_1, X_2 \in S_{++}^n$$

and let $\theta \in [0, 1]$.

Define

$$X_\theta = \theta X_1 + (1 - \theta) X_2.$$

Since X_1 and X_2 are symmetric, X_θ is also symmetric.

For any nonzero vector $\mathbf{z} \in \mathbb{R}^n$,

$$\mathbf{z}^\top X_\theta \mathbf{z} = \theta \mathbf{z}^\top X_1 \mathbf{z} + (1 - \theta) \mathbf{z}^\top X_2 \mathbf{z}.$$

Because $X_1 \succ 0$ and $X_2 \succ 0$,

$$\mathbf{z}^\top X_1 \mathbf{z} > 0, \quad \mathbf{z}^\top X_2 \mathbf{z} > 0.$$

Therefore,

$$\mathbf{z}^\top X_\theta \mathbf{z} > 0$$

for every $\mathbf{z} \neq 0$.

Thus

$$X_\theta \succ 0,$$

and hence

$$S_{++}^n \text{ is convex.}$$

2. Restriction to a line.

Fix $X \succ 0$ and a symmetric direction V .

Consider

$$g(t) = f(X + tV) = -\log \det(X + tV),$$

for all t such that

$$X + tV \succ 0.$$

To prove convexity of f , it is sufficient to show that its restriction to every feasible line is convex.

3. Factorization.

Since $X \succ 0$, both $X^{1/2}$ and $X^{-1/2}$ exist.

We have

$$X + tV = X^{1/2} \left(I + tX^{-1/2} V X^{-1/2} \right) X^{1/2}.$$

Indeed,

$$\begin{aligned} & X^{1/2} \left(I + tX^{-1/2} V X^{-1/2} \right) X^{1/2} \\ &= X + tX^{1/2} X^{-1/2} V X^{-1/2} X^{1/2} \\ &= X + tV. \end{aligned}$$

4. Expressing the determinant using eigenvalues.

Define

$$M = X^{-1/2} V X^{-1/2}.$$

Since $X^{-1/2}$ and V are symmetric, M is symmetric. Hence it has real eigenvalues

$$\lambda_1, \dots, \lambda_n.$$

Using

$$\det(ABC) = \det(A) \det(B) \det(C),$$

we obtain

$$\begin{aligned} \det(X + tV) &= \det(X^{1/2}) \det(I + tM) \det(X^{1/2}) \\ &= \det(X) \det(I + tM). \end{aligned}$$

Therefore,

$$\begin{aligned} g(t) &= -\log \det(X + tV) \\ &= -\log \det X - \log \det(I + tM). \end{aligned}$$

The eigenvalues of $I + tM$ are

$$1 + t\lambda_1, \dots, 1 + t\lambda_n.$$

Hence

$$\det(I + tM) = \prod_{i=1}^n (1 + t\lambda_i),$$

and therefore

$$g(t) = -\log \det X - \sum_{i=1}^n \log(1 + t\lambda_i).$$

5. Second derivative.

Differentiating once,

$$g'(t) = -\sum_{i=1}^n \frac{\lambda_i}{1 + t\lambda_i}.$$

Differentiating again,

$$g''(t) = \sum_{i=1}^n \frac{\lambda_i^2}{(1 + t\lambda_i)^2}.$$

Since

$$X + tV \succ 0,$$

we have

$$1 + t\lambda_i > 0 \quad \text{for every } i.$$

Thus every term satisfies

$$\frac{\lambda_i^2}{(1 + t\lambda_i)^2} \geq 0.$$

Hence

$$g''(t) \geq 0.$$

Therefore g is convex on every feasible line.

6. Conclusion.

Since the restriction of f to every line in its convex domain S_{++}^n is convex, we conclude that

$$f(X) = -\log \det X \text{ is convex on } S_{++}^n.$$